

Ecommerce Customer Churn Analysis

Welcome to my Customer Churn Analysis project. I tackled this project to expand my SQL knowledge and to better understand navigating databases. The goal was to analyze the data, understand customer behaviors, and provide insight and recommendations to reduce churn rates. The analysis is based on a dataset containing various customer attributes and behaviors. This repository also contains the queries I ran to get my results.

Project Approach

I got this dataset from Kaggle, and it contains a lot of different customer information. I used SQL to clean, explore, and analyze this dataset. This analysis has several sections: Data Cleaning, Data Exploration, Analysis and Insights, and Recommendations. Below I explain what each attribute found in the dataset means.

Attributes Explained

CustomerID: a unique ID for each customer

Churn: a value of 1 or 0 based on whether a customer (1 = churned, stayed = 0)

Tenure: how long a customer has/had been with the company (in months)

PreferredLoginDevice: what device a customer typically uses

CityTier: a value from 1 to 3 based on the type of city a customer lives in (1 = major metro, 2 = mid-size city, 3 = small city)

WarehouseToHome: Distance from the nearest warehouse to a customer's home (in km)

PreferredPaymentMode: what payment method a customer typically uses

Gender: a customer's gender

HourSpendOnApp: how many hours a week a customer spends on the app

NumberOfDeviceRegistered: how many devices a customer has registered to their account

PreferredOrderCat: what category a customer typically buys from

SatisfactionScore: a value from 1 to 5 chosen by a customer based on their satisfaction

MaritalStatus: a customer's marital status

NumberOfAddress: how many addresses a customer has saved

Complain: a value of 1 or 0 based on whether a customer has filed a complaint (1 = yes, 0 = no)

OrderAmountHikeFromLastYear: increase in a customer's order amount compared to last year

CouponUsed: how many coupons a customer has used

OrderCount: how many orders a customer has made

DaySinceLastOrder: how many days since a customer's last order

CashbackAmount: cashback a customer has received

Data Cleaning

Before I dive into the dataset and start exploring, it's important to make sure it's reliable and accurate. So, we need to clean it with several steps: handling missing/null values, checking for and correcting inconsistencies, and formatting the data for easier analysis. This will help make sure the integrity of the results.

1. Finding the total number of customers

```
SELECT COUNT(*) AS TotalCustomers
FROM customers;
```

There are 5,630 customers in this dataset.

2. Checking for duplicate rows

```
SELECT CustomerID, COUNT (CustomerID) AS Count
FROM customers
GROUP BY CustomerID
HAVING COUNT (CustomerID) > 1;
```

The query returns with an empty table, meaning there are no duplicate rows.

3. Checking for null values

We need to look at the columns with null values present.

```
SELECT 'Tenure' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE Tenure IS NULL
UNION
SELECT 'WarehouseToHome' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE WarehouseToHome IS NULL
UNION
```

```

SELECT 'HourSpendOnApp' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE HourSpendOnApp IS NULL
UNION
SELECT 'OrderAmountHikeFromLastYear' as ColumnName, COUNT(*) AS
NullCount
FROM customers WHERE OrderAmountHikeFromLastYear IS NULL
UNION
SELECT 'CouponUsed' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE CouponUsed IS NULL
UNION
SELECT 'OrderCount' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE OrderCount IS NULL
UNION
SELECT 'DaySinceLastOrder' as ColumnName, COUNT(*) AS NullCount
FROM customers WHERE DaySinceLastOrder IS NULL;

```

All of these columns have null values in them, and the amount of each of them is shown in the table above.

4. Filling null values with the mean from their attribute column

Ok, we need to deal with these null values. We do this by getting the mean value from each respective column and filling the null value in with it.

```

UPDATE customers
SET HourSpendOnApp = (SELECT AVG(HourSpendOnApp) FROM customers)
WHERE HourSpendOnApp IS NULL;

```

```

UPDATE customers
SET Tenure = (SELECT AVG(Tenure) FROM customers)
WHERE Tenure IS NULL;

```

```

UPDATE customers

```

```
SET OrderAmountHikeFromLastYear = (SELECT  
AVG(OrderAmountHikeFromLastYear) FROM customers)  
WHERE OrderAmountHikeFromLastYear IS NULL;
```

```
UPDATE customers  
SET WarehouseToHome = (SELECT AVG(WarehouseToHome) FROM customers)  
WHERE WarehouseToHome IS NULL;
```

```
UPDATE customers  
SET CouponUsed = (SELECT AVG(CouponUsed) FROM customers)  
WHERE CouponUsed IS NULL;
```

```
UPDATE customers  
SET OrderCount = (SELECT AVG(OrderCount) FROM customers)  
WHERE OrderCount IS NULL;
```

```
UPDATE customers  
SET DaySinceLastOrder = (SELECT AVG(DaySinceLastOrder) FROM  
customers)  
WHERE DaySinceLastOrder IS NULL;
```

The null values have been replaced, so now there are no more null values to be found.

5. Creating CustomerStatus attribute based off values of Churn attribute

The column 'Churn' contains values of 1 and 0. 1 meaning a customer churned and 0 meaning they stayed. To make the database table easy to follow I added an additional column called 'CustomerStatus' that shows "Stayed" or "Churned" based off of that 'Churn' column.

```
ALTER TABLE customers  
ADD CustomerStatus TEXT;
```

```
UPDATE customers  
SET CustomerStatus =
```

```
CASE
    WHEN Churn = 1 THEN 'Churned'
    WHEN Churn = 0 THEN 'Stayed'
END;
```

6. Creating ComplainReceived attribute based off values of Complain attribute

The same thing was done with the 'Complain' column. I added a column called 'ComplainReceived' so you can see "Yes" or "No" rather than a 1 or 0.

```
ALTER TABLE customers
ADD ComplainReceived TEXT;

UPDATE customers
SET ComplainReceived =
CASE
    WHEN Complain = 1 THEN 'Yes'
    WHEN Complain = 0 THEN 'No'
END;
```

7. Fix redundancy by setting Mobile Phone to Mobile

Some columns have redundant values that could be streamlined. These queries helped fix that issue.

```
UPDATE customers
SET PreferredLoginDevice = 'Mobile', PreferredOrderCat = 'Mobile'
WHERE PreferredLoginDevice = 'Mobile Phone'
OR PreferredOrderCat = 'Mobile Phone';
```

8. Fix redundancy by setting COD to Cash on Delivery

The same thing was done here.

```
UPDATE customers
SET PreferredPaymentMode = 'Cash on Delivery'
WHERE PreferredPaymentMode = 'COD';
```

9. Fixing incorrect values in WarehouseToHome

There were some outlier values that weren't correct in the WarehouseToHome column. These values were quickly fixed with these queries.

```
UPDATE customers
SET WarehouseToHome = '27'
WHERE WarehouseToHome = '127';
```

```
UPDATE customers
SET WarehouseToHome = '26'
WHERE WarehouseToHome = '126';
```

Data Exploration

This is where I explore the data and answer business questions.

1. What is the churn rate? How many customers churned and how many stayed?

```
SELECT TotalCustomers,
       ChurnedCustomers,
       CAST((ChurnedCustomers * 1.0 / TotalCustomers) * 100 AS
DECIMAL(10,2)) AS ChurnRate
FROM
  (SELECT COUNT(*) AS TotalCustomers FROM customers) AS Total,
  (SELECT COUNT(*) AS ChurnedCustomers FROM customers
WHERE CustomerStatus = 'Churned') AS Churned;
```

	TotalCustomers	ChurnedCustomers	ChurnRate
1	5630	948	16.8383658969805

The churn rate for this company is 16.84%. This means that out of the 5,630 customers, 948 churned. This is a big portion of the customer base that is leaving the company and needs to be addressed.

2. What is a churned customer's typical tenure?

First, I created a column called 'TenureRange' that groups different customers based on how long they've been a customer.

```
ALTER TABLE customers
ADD TenureRange TEXT;
```

```
UPDATE customers
SET TenureRange =
CASE
    WHEN ROUND(Tenure, 0) <= 6 THEN '0-6 Months'
    WHEN ROUND(Tenure, 0) > 6 AND ROUND(Tenure, 0) <= 12 THEN '6-12
Months'
    WHEN ROUND(Tenure, 0) > 12 AND ROUND(Tenure, 0) <= 24 THEN '1-2
Years'
    WHEN ROUND(Tenure, 0) > 24 THEN '2+ Years'
END;
```

```
SELECT TenureRange,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY TenureRange
ORDER BY ChurnRate DESC;
```

	TenureRange	TotalCustomers	ChurnedCustomers	ChurnRate
1	0-6 Months	2150	697	32.4186046511628
2	6-12 Months	1584	156	9.84848484848485
3	1-2 Years	1467	95	6.47580095432856
4	2+ Years	429	0	0.0

697 of the total 948 churned customers churned within 0-6 months. Customers who are with the company for longer periods tend to churn less, especially the group of 2+ years.

3. What does churn rate look like depending on preferred login device?

```
SELECT PreferredLoginDevice,
       COUNT(*) AS TotalCustomers,
       SUM(churn) AS ChurnedCustomers,
       CAST(SUM (churn) * 1.0 / COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY PreferredLoginDevice;
```

	PreferredLoginDevice	TotalCustomers	ChurnedCustomers	ChurnRate
1	Computer	1634	324	19.828641370869
2	Mobile	3996	624	15.6156156156156

Customers who prefer to login on computer have a higher rate of churn compared to customers who login on mobile. Customers on computers may have different user experiences or patterns that may contribute to higher churn than those on mobile.

4. What does churn rate look like depending on city tier?

```
SELECT CityTier,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM (churn) * 1.0 / COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
```



```
FROM customers
GROUP BY CityTier
ORDER BY ChurnRate DESC;
```

	CityTier	TotalCustomers	ChurnedCustomers	ChurnRate
1	3	1722	368	21.3704994192799
2	2	242	48	19.8347107438017
3	1	3666	532	14.5117294053464

City tier 3 has the highest churn rate at 21.4%, city tier 2 follows with 19.8%, and city tier 1 ends with 14.5%. These results show that city tier may have an effect on churn rates with customers. City tier 1 has the lowest churn rate which could be due to customer preference or lack of alternatives.

5. What does churn rate look like depending on a customer's distance from nearest warehouse?

Here I created another column called 'WarehouseDistanceRange' that groups different distances of warehouses from customers' homes.

```
ALTER TABLE customers
ADD WarehouseDistanceRange TEXT;

UPDATE customers
SET WarehouseDistanceRange =
CASE
    WHEN WarehouseToHome <= 10 THEN '0-10 km'
    WHEN WarehouseToHome > 10 AND WarehouseToHome <= 20 THEN '10-20
km'
    WHEN WarehouseToHome > 20 AND WarehouseToHome <= 30 THEN '20-30
km'
    WHEN WarehouseToHome > 30 THEN '30+ km'
END;

SELECT WarehouseDistanceRange,
```

```

COUNT(*) AS TotalCustomers,
SUM(Churn) AS ChurnedCustomers,
CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY WarehouseDistanceRange
ORDER BY ChurnRate DESC;

```

	WarehouseDistanceRange	TotalCustomers	ChurnedCustomers	ChurnRate
1	30+ km	469	98	20.8955223880597
2	20-30 km	874	176	20.137299771167
3	10-20 km	2318	408	17.6013805004314
4	0-10 km	1969	266	13.5093956323007

The distance between customers' homes and warehouses seems to have a slight effect on churn rates. The churn rate increases as a customer's distance from a warehouse increases. These rates could be due to factors such as delivery times/costs or overall customer convenience. If inconvenienced, a customer is less likely to stay with the company.

6. What does churn rate look like depending on preferred payment method?

```

SELECT PreferredPaymentMode,
COUNT(*) AS TotalCustomers,
SUM(Churn) AS ChurnedCustomers,
CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY PreferredPaymentMode
ORDER BY ChurnRate DESC;

```

	PreferredPaymentMode	TotalCustomers	ChurnedCustomers	ChurnRate
1	Cash on Delivery	514	128	24.9027237354086
2	E wallet	614	140	22.8013029315961
3	CC	273	59	21.6117216117216
4	UPI	414	72	17.3913043478261
5	Debit Card	2314	356	15.3846153846154
6	Credit Card	1501	193	12.8580946035976

The preferred payment method of customers seems to have some influence on churn rate. Customers who use cash on delivery have the highest churn rate at 24.9% while debit and credit cards have the lowest churn rates.

7. Are men or women churning more?

```
SELECT Gender,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY Gender
ORDER BY ChurnRate DESC;
```

	Gender	TotalCustomers	ChurnedCustomers	ChurnRate
1	Male	3384	600	17.7304964539007
2	Female	2246	348	15.4942119323241

Men churn slightly more at 17.7% than women at 15.5%. However, because these rates are so close, gender doesn't seem to influence churn.

8. Do churned customers spend less time on the app?

```
SELECT CustomerStatus,
```

```

ROUND (AVG (HourSpendOnApp), 2) AS AvgHoursOnApp
FROM customers
GROUP BY CustomerStatus;

```

	CustomerStatus	AvgHoursOnApp
1	Churned	2.96
2	Stayed	2.93

Both churning and staying customers roughly average 3 hours weekly on the app. This shows that the average weekly app usage does not affect churn.

9. What does churn rate look like depending on number of registered devices?

```

SELECT NumberofDeviceRegistered,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY NumberofDeviceRegistered
ORDER BY ChurnRate DESC;

```

	NumberOfDeviceRegistered	TotalCustomers	ChurnedCustomers	ChurnRate
1	6	162	56	34.5679012345679
2	5	881	198	22.4744608399546
3	4	2377	392	16.4913756836348
4	3	1699	254	14.9499705709241
5	2	276	26	9.42028985507246
6	1	235	22	9.36170212765957

There does seem to be a correlation between number of devices registered and churn rate. As the number of devices registered increases so does customer churn. The largest number of devices registered is 6 and the churn rate for those customers is 34.6%. On the other end with 1 device registered, the churn rate is 9.4%.

10. What does churn rate look like depending on preferred order category?

```
SELECT PreferredOrderCat,  
       COUNT(*) AS TotalCustomers,  
       SUM(Churn) AS ChurnedCustomers,  
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS  
ChurnRate  
FROM customers  
GROUP BY PreferredOrderCat  
ORDER BY ChurnRate DESC;
```

	PreferredOrderCat	TotalCustomers	ChurnedCustomers	ChurnRate
1	Mobile	2080	570	27.4038461538462
2	Fashion	826	128	15.4963680387409
3	Laptop & Accessory	2050	210	10.2439024390244
4	Others	264	20	7.57575757575758
5	Grocery	410	20	4.8780487804878

It looks like a customer's preferred shopping category may affect churn. The 'Mobile' category has the highest churn rate at 27.4% while the 'Grocery' category has the lowest churn rate at 4.9%. Mobile category shoppers may need better retention strategies while grocery category shoppers appear loyal.

11. What are the churn rates for each satisfaction level?

```
SELECT SatisfactionScore,  
       COUNT(*) AS TotalCustomers,  
       SUM(Churn) AS ChurnedCustomers,  
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS  
ChurnRate  
FROM customers  
GROUP BY SatisfactionScore  
ORDER BY ChurnRate DESC;
```

	SatisfactionScore	TotalCustomers	ChurnedCustomers	ChurnRate
1	5	1108	264	23.826714801444
2	3	1698	292	17.1967020023557
3	4	1074	184	17.1322160148976
4	2	586	74	12.6279863481229
5	1	1164	134	11.5120274914089

Satisfaction scores do have an effect on churn rates but not in the way you would expect. The lower the satisfaction score, the lower the churn rate. This shows that even though customers may appear to be satisfied, they can still churn just as easily if not more than anyone else.

12. What does churn rate look like depending on marital status?

```
SELECT MaritalStatus,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY MaritalStatus
ORDER BY ChurnRate DESC;
```

	MaritalStatus	TotalCustomers	ChurnedCustomers	ChurnRate
1	Single	1796	480	26.7260579064588
2	Divorced	848	124	14.622641509434
3	Married	2986	344	11.5204286671132

Out of the marital statuses, single customers churn the most at 26.7%. Divorced customers churn at 14.6% and married customers churn the lowest at 11.5%.

13. Does average number of addresses affect churn rate?

```

SELECT
    CustomerStatus,
    ROUND (AVG (NumberOfAddress), 2) AS AvgAddressCount
FROM customers
GROUP BY CustomerStatus
ORDER BY CustomerStatus;

```

	CustomerStatus	AvgAddressCount
1	Churned	4.47
2	Stayed	4.16

On average, both churning and staying customers have 4 addresses. This shows that the amount of addresses doesn't affect churn rate.

14. What does churn rate look like depending on complaints?

```

SELECT ComplainReceived,
    COUNT(*) AS TotalCustomers,
    SUM(Churn) AS ChurnedCustomers,
    CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY ComplainReceived
ORDER BY ChurnRate DESC;

```

	ComplainReceived	TotalCustomers	ChurnedCustomers	ChurnRate
1	Yes	1604	508	31.6708229426434
2	No	4026	440	10.9289617486339

Customers who complained had a 31.7% churn rate while customers who didn't complain had a 10.9% churn rate. This is a huge discovery in this dataset showing the importance of keeping customers happy and loyal. There is a big gap between these 2 churn rates that need to be addressed to decrease the amount of customers leaving. Listening to and addressing customer

feedback is one way to do this. Proactivity is also important so that customers don't complain in the first place. Consistently improving the overall customer experience goes a long way in retaining customers and building strong relationships.

15. What does coupon usage tell us?

```
SELECT CustomerStatus, SUM(CouponUsed) AS SumOfCouponUsed
FROM customers
GROUP BY CustomerStatus;
```

	CustomerStatus	SumOfCouponUsed
1	Churned	1628.00818756978
2	Stayed	8230.25381466319

Customers who stay use substantially more coupons than customers who churn. Customers who use coupons show more engagement and loyalty. It's important to cross that over to more customers so they also engage and stay. Personalizing offers and emphasizing loyalty rewards can help with coupon engagement and thus help customer retention.

16. How does cashback influence churn rate?

For this question I needed to add another column called 'CashbackAmountRange' that groups different cashback amounts into neat ranges.

```
ALTER TABLE customers
ADD CashbackAmountRange TEXT;

UPDATE customers
SET CashbackAmountRange =
CASE
    WHEN CashbackAmount <= 100 THEN 'Low Cashback Amount'
    WHEN CashbackAmount > 100 AND CashbackAmount <= 200 THEN 'Moderate
Cashback Amount'
    WHEN CashbackAmount > 200 AND CashbackAmount <= 300 THEN 'High
Cashback Amount'
    WHEN CashbackAmount > 300 THEN 'Very High Cashback Amount'
```


END;

```
SELECT CashbackAmountRange,
       COUNT(*) AS TotalCustomers,
       SUM(Churn) AS ChurnedCustomers,
       CAST(SUM(Churn) * 1.0 /COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
GROUP BY CashbackAmountRange
ORDER BY ChurnRate DESC;
```

	CashbackAmountRange	TotalCustomers	ChurnedCustomers	ChurnRate
1	Moderate Cashback Amount	4305	814	18.9082462253194
2	High Cashback Amount	1157	124	10.7173725151253
3	Very High Cashback Amount	156	10	6.41025641025641
4	Low Cashback Amount	12	0	0.0

Customers with a moderate cashback amount had the highest churn rate of 18.9%, followed by high cashback amount at 10.7%, then very high cashback amount at 6.4%, and lastly low cashback amount at 0%. These rates suggest that offering higher cashback amounts can help reduce customer churn.

17. Average order count?

```
SELECT
    CustomerStatus,
    ROUND(AVG(OrderCount), 2) AS AvgOrderCount
FROM customers
GROUP BY CustomerStatus
ORDER BY CustomerStatus;
```

	CustomerStatus	AvgOrderCount
1	Churned	2.83
2	Stayed	3.04

The average order count for churning and staying customers was about 3. So, churn is not caused by how many orders a customer makes.

18. What's the average days since last order for churned and staying customers?

```
SELECT
    CustomerStatus,
    ROUND(AVG(DaySinceLastOrder), 2) AS AvgDaySinceLastOrder
FROM customers
GROUP BY CustomerStatus
ORDER BY CustomerStatus;
```

	CustomerStatus	AvgDaySinceLastOrder
1	Churned	3.31
2	Stayed	4.79

The average days since last order for churned customers was about 3.3. For staying customers it was about 4.8. Churned customers may have had a recent purchase that was negative and decided to leave.

19. How many people churned with just 1 purchase?

```
SELECT
    OrderCount,
    COUNT(*) AS TotalCustomers,
    SUM(Churn) AS ChurnedCustomers,
    CAST(SUM(Churn) * 1.0 / COUNT(*) * 100 AS DECIMAL(10,2)) AS
ChurnRate
FROM customers
WHERE OrderCount = 1
GROUP BY OrderCount;
```

	OrderCount	TotalCustomers	ChurnedCustomers	ChurnRate
1	1	1751	316	18.0468303826385

1,751 customers made 1 purchase, and of those customers 316 churned. This gives us a churn rate of 18%. One-time buyers churn at a lower rate than the 32% churn rate in the 0-6 month group. This means that not all early churners are one-time buyers.

Analysis and Insights

- The overall churn is 16.84%, with 73.5% of all churn happening in the first 0-6 months. This shows that early-stage onboarding and first-purchase experience are big retention weaknesses.
- Customers who stay longer than 2 years almost never churn, indicating strong loyalty once habits are formed.
- One-purchase customers churn at 18%. This is lower than the 32% churn rate in the 0-6 month group, meaning not all early churners are one-time buyers.
- Customers who prefer logging in on computer churn more than mobile users. This could point to UX differences or friction points on desktop.
- Lower satisfaction scores correlate with *lower* churn. This suggests that satisfaction may be unreliable or customers may churn without expressing dissatisfaction.
- Churn rate is the highest in Tier 3 (21.4%) and lowest in Tier 1 (14.5%). This indicates that customers in smaller markets may have less alternatives or lower satisfaction.
- Churn rises as distance from the warehouse increases, likely due to slower delivery times or higher inconvenience.
- Cash on delivery customers churn at 24.9% which is far higher than card users. This suggests that COD customers may be less committed or more sensitive to friction.
- Single customers churn the most at 26.7%, while married customers churn the least at 11.5%. This could suggest that stability correlates with retention.
- App usage doesn't predict churn among customers.
- More registered devices equals higher churn, indicating multi-device users are less loyal or more experimental.
- Mobile category shoppers churn the most (27.4%), while Grocery category shoppers churn the least (4.9%). This shows category-specific retention needs.

- Customers who complain churn at 31.7% while customers who don't churn at 10.9%. This is one of the strongest churn indicators in the dataset.
- Staying customers use significantly more coupons, showing that coupon engagement is a strong loyalty signal.
- Higher cashback amounts correlate with lower churn. This suggests that cashback is a good retention method.
- Churn is not driven by how many orders a customer makes.
- Customers who churn often make a recent purchase before leaving. This could mean that their last interaction was either negative or disappointing.

Recommendations

1. Improve early-stage onboarding - 73.5% of churn happens in the first 0-6 months. Focusing on strengthening onboarding as well as first-purchase experience will reduce churn more than any other action.
2. Reduce complaint-related churn - Customers who complain churn at 31.7%. This is nearly triple non-complainers. Proactive support and better communication can significantly reduce churn.
3. Enhance desktop user experience - Computer users churn more than mobile users. If there is friction on the desktop side of the app, then improving UI and checkout flow can reduce churn.
4. Strengthen retention in high-risk city tiers - Better delivery options, localized offers, or targeted marketing are just some ideas that could help retain customers in smaller markets.
5. Encourage cashback and coupon engagement - Since coupon users stay longer and higher cashback amounts reduce churn, targeted promotions and loyalty rewards can increase retention.
6. Improve delivery experience for distant customers - Churn rates increase as warehouse distance increases. Faster deliveries and reduced fees could help this problem.
7. Focus retention efforts on high-risk categories - Mobile category shoppers churn the highest (27.4%). Category specific bundles, discounts, or personalized offers can help.

By acting on these insights and recommendations, the company can reduce key drivers of churn and improve overall customer satisfaction. These changes will help retain more customers, increase long-term loyalty, and support sustainable growth across all segments.